Darius Foo

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Skills

Formal verification, higher-order concurrent separation logic, Iris; Lean, Rust, OCaml, TLA$^+$

Education

National University of Singapore

PhD in Computer Science, under Graduate Tutorship-PhD Scheme — *2020 – 2026*

- Thesis: *Automated Verification of Effectful Higher-order Programs*
- Dean's Graduate Research Excellence Award, AY 2023/2024

National University of Singapore

Bachelor of Computing (Honours) in Computer Science — *2012 – 2016*

Publications

Iris on Indirection Theory: Lightweight Higher-order Ghost State — Under submission, 2026
Darius Foo, Viet-Anh Nguyen, Wei-Ngan Chin

A Program Logic for Contextual Refinement of Imperative Shift/Reset Programs — Under submission, 2026
Darius Foo, Viet-Anh Nguyen, Wei-Ngan Chin

Specifying and Verifying Future Conditions — SAS 2025
Yahui Song, Darius Foo, Wei-Ngan Chin

Specification and Verification for Unrestricted Algebraic Effects and Handling — ICFP 2024
Yahui Song, Darius Foo, Wei-Ngan Chin

Staged Specification Logic for Verifying Higher-Order Imperative Programs — FM 2024
Darius Foo, Yahui Song, Wei-Ngan Chin

Protocol Conformance with Choreographic PlusCal — TASE 2023
Darius Foo, Andreea Costea, Wei-Ngan Chin

Automated Temporal Verification for Higher-order Algebraic Effects — APLAS 2022
Yahui Song, Darius Foo, Wei-Ngan Chin

Tracing OCaml Programs — OCaml 2022
Darius Foo, Wei-Ngan Chin

Work Experience

Research Fellow — *2026 – Present*
National University of Singapore

Senior Software Engineer — *2016 – 2020*
SourceClear, later Veracode by acquisition

Worked on the SourceClear Agent, a tool which uses a combination of static analysis and instrumentation to discover and automatically upgrade library dependencies from CI/CD pipelines.

Academic Service

Programme Committee	OCaml 2025
Subreviewer	CAV 2022, POPL 2022, TASE 2024
Artifact Evaluation Committee	ESOP 2022
Student Volunteer	PLDI 2021, SPLASH 2022

Teaching

Database Systems	2020, 2022, 2024 S1 & S2, 2025
Programming Methodology I & II	2013-2015, 2021-2023
Software Engineering	2016, 2022